MS-01 Soil Resistivity – Measurement Sheet

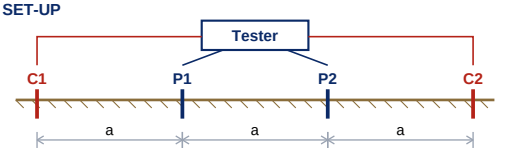
Wenner 4-probe · IS 3043:2018 Cl. 40 · IEEE 81-2012 Cl. 7

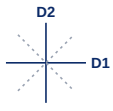
Client Gujarat Fluorochemicals Ltd	**Project No.** 024-D03_TS-26-GFL	**Work Order No.** 1002-1261201769
Site / Unit Dahej – FI Plant	**Work date** ____ / ____ / 20____	**Sheet** ____ of ____
ETPL engineer(s)	**Client representative**	**Ground condition** ☐ Dry ☐ Damp ☐ Wet
Location name	**Reading unit** ☐ R (Ω) ☐ ρ (Ω·m)	**Probe depth (cm)**
Direction 1 / 2 bearing D1 _____° D2 _____°	**Direction 3 / 4 bearing** D3 _____° D4 _____°	**Weather** ☐ Sunny ☐ Cloudy ☐ After rain

Instrument used: ☐ Metrel MI 3290 GX Earth Analyser ☐ Kyoritsu 4105A Digital Earth Tester Sr. No. _____ Cal. due ___/___/____

Sr.	Probe spacing a (m)	Direction 1 reading	Direction 2 reading	Direction 3 reading	Direction 4 reading	Remark
1	0.50					
2	1.00					
3	1.50					
4	2.00					
5	2.50					
6	3.00					
7	3.50					
8	4.00					
9	4.50					
10	5.00					

SET-UP





D3/D4 dashed

Equal spacing a · 10–15 cm deep · I via C1–C2, V across P1–P2 · ρ = 2πaR

HOW TO RECORD

- Write the value shown on the instrument and tick its unit in the header: R (Ω) or ρ (Ω·m). The app calculates ρ = 2πaR, mean per spacing, average and IS 3043 Table 4 corrosivity.
- Four probes in a straight line at equal spacing a, driven 10–15 cm. If depth exceeds a/20 the app uses the full formula (Cl. 40.4.1).
- Substations / generating stations: at least 8 directions from the centre (Cl. 40.2.1) – use extra sheets. Keep the line clear of buried metal and earth conductors.

ETPL engineer – name & signature	Client representative – name & signature	Time start / finish

MS-02 Earth Electrode (Pit) Resistance – Measurement Sheet

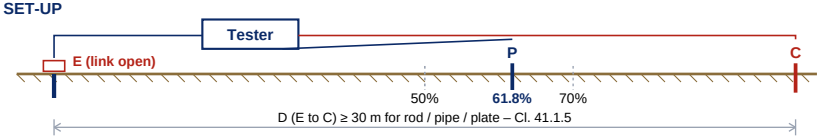
Fall of potential 61.8% · IS 3043:2018 Cl. 14.2, 41 · IEEE 81-2012 Cl. 8

Client Gujarat Fluorochemicals Ltd	**Project No.** 024-D03_TS-26-GFL	**Work Order No.** 1002-1261201769
Site / Unit Dahej – FI Plant	**Work date** ____ / ____ / 20____	**Sheet** ____ of ____
ETPL engineer(s)	**Client representative**	**Ground condition** ☐ Dry ☐ Damp ☐ Wet
Current electrode distance C (m)	**Auxiliary rods** ☐ 12.5 mm MS, ≤ 1 m deep	**90°/180° check done** ☐ Yes ☐ No

Instrument used: ☐ Metrel MI 3290 GX Earth Analyser ☐ Kyoritsu 4105A Digital Earth Tester Sr. No. _____ Cal. due ___/___/____

Sr.	Pit TAG No.	Location / name of pit	Electrode type	Material	Length l (m)	Dia d / strip width (mm)	R without grid at 61.8% (Ω)	Check R at 50% (Ω)	Check R at 70% (Ω)	R with grid (Ω)	Method	Condition codes	Photo ref.	Remark
1	GFL-FI-EP-01	Transformer TR-1	☐ Plate ☐ Pipe ☐ Rod ☐ Strip	☐ Cu ☐ Cu-bonded ☐ GI ☐ MS ☐ CI ☐ SS							☐ 61.8% ☐ Clamp ☐ Slope	☐ C1 ☐ C2 ☐ C3 ☐ C4 ☐ C5 ☐ C6 ☐ C7 ☐ C8		
2	GFL-FI-EP-02	Transformer TR-1	☐ Plate ☐ Pipe ☐ Rod ☐ Strip	☐ Cu ☐ Cu-bonded ☐ GI ☐ MS ☐ CI ☐ SS							☐ 61.8% ☐ Clamp ☐ Slope	☐ C1 ☐ C2 ☐ C3 ☐ C4 ☐ C5 ☐ C6 ☐ C7 ☐ C8		
3	GFL-FI-EP-03	MCC Room	☐ Plate ☐ Pipe ☐ Rod ☐ Strip	☐ Cu ☐ Cu-bonded ☐ GI ☐ MS ☐ CI ☐ SS							☐ 61.8% ☐ Clamp ☐ Slope	☐ C1 ☐ C2 ☐ C3 ☐ C4 ☐ C5 ☐ C6 ☐ C7 ☐ C8		
4	GFL-FI-EP-04	MCC Room	☐ Plate ☐ Pipe ☐ Rod ☐ Strip	☐ Cu ☐ Cu-bonded ☐ GI ☐ MS ☐ CI ☐ SS							☐ 61.8% ☐ Clamp ☐ Slope	☐ C1 ☐ C2 ☐ C3 ☐ C4 ☐ C5 ☐ C6 ☐ C7 ☐ C8		
5	GFL-FI-EP-05	Reactor Bay R-201	☐ Plate ☐ Pipe ☐ Rod ☐ Strip	☐ Cu ☐ Cu-bonded ☐ GI ☐ MS ☐ CI ☐ SS							☐ 61.8% ☐ Clamp ☐ Slope	☐ C1 ☐ C2 ☐ C3 ☐ C4 ☐ C5 ☐ C6 ☐ C7 ☐ C8		
6	GFL-FI-EP-06	Reactor Bay R-202	☐ Plate ☐ Pipe ☐ Rod ☐ Strip	☐ Cu ☐ Cu-bonded ☐ GI ☐ MS ☐ CI ☐ SS							☐ 61.8% ☐ Clamp ☐ Slope	☐ C1 ☐ C2 ☐ C3 ☐ C4 ☐ C5 ☐ C6 ☐ C7 ☐ C8		
7	GFL-FI-EP-07	Cooling Tower	☐ Plate ☐ Pipe ☐ Rod ☐ Strip	☐ Cu ☐ Cu-bonded ☐ GI ☐ MS ☐ CI ☐ SS							☐ 61.8% ☐ Clamp ☐ Slope	☐ C1 ☐ C2 ☐ C3 ☐ C4 ☐ C5 ☐ C6 ☐ C7 ☐ C8		
8	GFL-FI-EP-08	Pump House	☐ Plate ☐ Pipe ☐ Rod ☐ Strip	☐ Cu ☐ Cu-bonded ☐ GI ☐ MS ☐ CI ☐ SS							☐ 61.8% ☐ Clamp ☐ Slope	☐ C1 ☐ C2 ☐ C3 ☐ C4 ☐ C5 ☐ C6 ☐ C7 ☐ C8		
9	GFL-FI-EP-09	Pipe Rack PR-3	☐ Plate ☐ Pipe ☐ Rod ☐ Strip	☐ Cu ☐ Cu-bonded ☐ GI ☐ MS ☐ CI ☐ SS							☐ 61.8% ☐ Clamp ☐ Slope	☐ C1 ☐ C2 ☐ C3 ☐ C4 ☐ C5 ☐ C6 ☐ C7 ☐ C8		
10	GFL-FI-EP-10	DG Area	☐ Plate ☐ Pipe ☐ Rod ☐ Strip	☐ Cu ☐ Cu-bonded ☐ GI ☐ MS ☐ CI ☐ SS							☐ 61.8% ☐ Clamp ☐ Slope	☐ C1 ☐ C2 ☐ C3 ☐ C4 ☐ C5 ☐ C6 ☐ C7 ☐ C8		

SET-UP



Read at 61.8%, confirm at 50% and 70% – values must agree (IS 3043:2018 Cl. 41.1.6)

HOW TO RECORD

- Open the test link for 'R without grid'; restore it after. Read at 61.8%, then move P to 50% and 70%: if the three readings agree the 61.8% value is valid (Cl. 41.1.6).
- Current electrode C at least 30 m from a rod, pipe or plate (Cl. 41.1.5); auxiliary rods 12.5 mm MS, up to 1 m deep (Cl. 41.1.4). Record C distance in the header.
- Plate: write plate size (m × m) in Remark. Pre-printed tags come from the asset register; new pits on a continuation sheet.

Condition codes: C1 Corroded · C2 Strip / test link disconnected · C3 Single earth connection (optional) · C4 Buried / chamber not accessible or damaged · C5 Electrode damaged · C6 Loose connection · C7 Joint not cleaned / tinned / protected · C8 No tag / label · OTH Other – write remark

ETPL engineer – name & signature	Client representative – name & signature	Time start / finish

MS-03 Earth Continuity & Bonding – Measurement Sheet

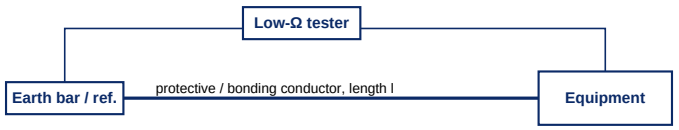
Ground integrity · IEEE 81-2012 Cl. 10 · IS 3043:2018 Cl. 17.2.3

Client Gujarat Fluorochemicals Ltd	**Project No.** 024-D03_TS-26-GFL	**Work Order No.** 1002-1261201769
Site / Unit Dahej – FI Plant	**Work date** ____ / ____ / 20____	**Sheet** ____ of ____
ETPL engineer(s)	**Client representative**	**Ground condition** ☐ Dry ☐ Damp ☐ Wet
Test-lead resistance (Ω)	**Lead null done** ☐ Yes	**Method** ☐ Low-Ω ☐ High-current

Instrument used: ☐ Metrel MI 3152 EurotestXC ☐ Metrel MI 3290 GX Earth Analyser Sr. No. _____ Cal. due ___/___/____

Sr.	TAG No.	Main location / area	Reference point (earth bar / riser)	Tested point (equipment / riser)	Distance (m)	Test current (A)	Measured R (Ω)	Condition codes	Photo ref.	Remark
1								☐ C1 ☐ C2 ☐ C6 ☐ C7		
2								☐ C1 ☐ C2 ☐ C6 ☐ C7		
3								☐ C1 ☐ C2 ☐ C6 ☐ C7		
4								☐ C1 ☐ C2 ☐ C6 ☐ C7		
5								☐ C1 ☐ C2 ☐ C6 ☐ C7		
6								☐ C1 ☐ C2 ☐ C6 ☐ C7		
7								☐ C1 ☐ C2 ☐ C6 ☐ C7		
8								☐ C1 ☐ C2 ☐ C6 ☐ C7		
9								☐ C1 ☐ C2 ☐ C6 ☐ C7		
10								☐ C1 ☐ C2 ☐ C6 ☐ C7		
11								☐ C1 ☐ C2 ☐ C6 ☐ C7		
12								☐ C1 ☐ C2 ☐ C6 ☐ C7		
13								☐ C1 ☐ C2 ☐ C6 ☐ C7		
14								☐ C1 ☐ C2 ☐ C6 ☐ C7		
15								☐ C1 ☐ C2 ☐ C6 ☐ C7		
16								☐ C1 ☐ C2 ☐ C6 ☐ C7		
17								☐ C1 ☐ C2 ☐ C6 ☐ C7		
18								☐ C1 ☐ C2 ☐ C6 ☐ C7		
19								☐ C1 ☐ C2 ☐ C6 ☐ C7		

SET-UP



1. Short the leads and null (record lead R) 2. Clean paint/rust at both points
3. Read R between reference and tested point (IEEE 81-2012 Cl. 10)

HOW TO RECORD

- Short the leads and null the tester; write lead R in the header – it is subtracted from every reading (IEEE 81-2012 Cl. 10.2).
- Keep one reference point for the area so results can be compared point to point. Record distance between the two points where known.
- High-current integrity test: write test current and use Remark for the voltage drop (V).

Condition codes: C1 Corroded · C2 Strip / test link disconnected · C3 Single earth connection (optional) · C4 Buried / chamber not accessible or damaged · C5 Electrode damaged C6 Loose connection · C7 Joint not cleaned / tinned / protected · C8 No tag / label · OTH Other – write remark

ETPL engineer – name & signature	Client representative – name & signature	Time start / finish

MS-04 Earth Fault Loop Impedance – Measurement Sheet

Zs, L-PE · IS 3043:2018 Cl. 42 · IS/IEC 60364-6

Client Gujarat Fluorochemicals Ltd	**Project No.** 024-D03_TS-26-GFL	**Work Order No.** 1002-1261201769
Site / Unit Dahej – FI Plant	**Work date** ____ / ____ / 20____	**Sheet** ____ of ____
ETPL engineer(s)	**Client representative**	**Ground condition** ☐ Dry ☐ Damp ☐ Wet

Instrument used: ☐ Kyoritsu 4118A Loop / PSC Tester ☐ Metrel MI 3152 EurotestXC Sr. No. _____ Cal. due ___/___/____

Sr.	Location of panel / equipment	Feeder name & tag no.	Device type	Device rating In (A)	Trip curve	Measured point	Loop impedance Zs (Ω)	Voltage L-PE (V)	Photo ref.	Remark
1			☐ ACB ☐ MCCB ☐ MCB ☐ SFU ☐ Fuse		☐ B ☐ C ☐ D ☐ Adj.	R-E				
						Y-E				
						B-E				
2			☐ ACB ☐ MCCB ☐ MCB ☐ SFU ☐ Fuse		☐ B ☐ C ☐ D ☐ Adj.	R-E				
						Y-E				
						B-E				
3			☐ ACB ☐ MCCB ☐ MCB ☐ SFU ☐ Fuse		☐ B ☐ C ☐ D ☐ Adj.	R-E				
						Y-E				
						B-E				
4			☐ ACB ☐ MCCB ☐ MCB ☐ SFU ☐ Fuse		☐ B ☐ C ☐ D ☐ Adj.	R-E				
						Y-E				
						B-E				

SET-UP



Zs = whole fault loop: phase → fault → PE → source (IS 3043:2018 Cl. 42)

HOW TO RECORD

- Live working: risk assessment, PPE and permit mandatory. Measure at the outgoing side of the device, one reading per phase to PE.
- Copy device type, rating In and trip curve from the nameplate – the app uses them for the device's disconnection limit.
- A panel is healthy only if all three phases meet the project's Zs criterion (Standards Acceptance List).

Condition codes: C1 Corroded · C2 Strip / test link disconnected · C3 Single earth connection (optional) · C4 Buried / chamber not accessible or damaged · C5 Electrode damaged C6 Loose connection · C7 Joint not cleaned / tinned / protected · C8 No tag / label · OTH Other – write remark

ETPL engineer – name & signature	Client representative – name & signature	Time start / finish

MS-05 Prospective Fault & Short-circuit Current – Measurement Sheet

PFC (L-PE) & PSC (L-N) · IS/IEC 60364-4-43 Reg. 434.5.1

Client Gujarat Fluorochemicals Ltd	**Project No.** 024-D03_TS-26-GFL	**Work Order No.** 1002-1261201769
Site / Unit Dahej – FI Plant	**Work date** ____ / ____ / 20____	**Sheet** ____ of ____
ETPL engineer(s)	**Client representative**	**Ground condition** ☐ Dry ☐ Damp ☐ Wet

Instrument used: ☐ Kyoritsu 4118A Loop / PSC Tester ☐ Metrel MI 3152 EurotestXC Sr. No. _____ Cal. due ___/___/____

Sr.	Location of panel / equipment	Feeder name & tag no.	Device type	Rating In (A)	Breaking capacity Icu / Icn (kA)	Phase	PFC L-PE (kA)	PSC L-N (kA)	Voltage L-N (V)	Photo ref.	Remark
1			☐ ACB ☐ MCCB ☐ MCB ☐ SFU ☐ Fuse			R					
						Y					
						B					
2			☐ ACB ☐ MCCB ☐ MCB ☐ SFU ☐ Fuse			R					
						Y					
						B					
3			☐ ACB ☐ MCCB ☐ MCB ☐ SFU ☐ Fuse			R					
						Y					
						B					
4			☐ ACB ☐ MCCB ☐ MCB ☐ SFU ☐ Fuse			R					
						Y					
						B					

SET-UP



L-PE → PFC · L-N → PSC · device breaking capacity (kA) must exceed the higher

HOW TO RECORD

- Breaking capacity (kA) from the nameplate is REQUIRED – without it the result is 'Not Assessable'.
- Record both PFC (L-PE) and PSC (L-N) per phase; the device must interrupt the higher of the two.
- Write values in kA as shown on the 4118A (convert A to kA if the display shows A).

Condition codes: C1 Corroded · C2 Strip / test link disconnected · C3 Single earth connection (optional) · C4 Buried / chamber not accessible or damaged · C5 Electrode damaged
C6 Loose connection · C7 Joint not cleaned / tinned / protected · C8 No tag / label · OTH Other – write remark

ETPL engineer – name & signature	Client representative – name & signature	Time start / finish

MS-06 Riser Integrity – Measurement Sheet

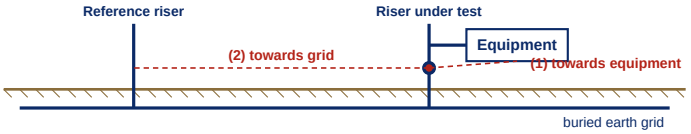
Two-direction continuity · IEEE 81-2012 Cl. 10

Client Gujarat Fluorochemicals Ltd	**Project No.** 024-D03_TS-26-GFL	**Work Order No.** 1002-1261201769
Site / Unit Dahej – FI Plant	**Work date** ____ / ____ / 20____	**Sheet** ____ of ____
ETPL engineer(s)	**Client representative**	**Ground condition** ☐ Dry ☐ Damp ☐ Wet
Test-lead resistance (Ω)	**Reference riser / point**	**Test current** ☐ 200 mA ☐ Other ____

Instrument used: ☐ Metrel MI 3290 GX Earth Analyser ☐ Metrel MI 3152 EurotestXC Sr. No. _____ Cal. due ___/___/____

Sr.	Main location	Equipment / riser / measurement point	R towards equipment (Ω)	R towards grid (Ω)	Condition codes	Photo ref.	Remark
1					☐ C1 ☐ C2 ☐ C6 ☐ C7		
2					☐ C1 ☐ C2 ☐ C6 ☐ C7		
3					☐ C1 ☐ C2 ☐ C6 ☐ C7		
4					☐ C1 ☐ C2 ☐ C6 ☐ C7		
5					☐ C1 ☐ C2 ☐ C6 ☐ C7		
6					☐ C1 ☐ C2 ☐ C6 ☐ C7		
7					☐ C1 ☐ C2 ☐ C6 ☐ C7		
8					☐ C1 ☐ C2 ☐ C6 ☐ C7		
9					☐ C1 ☐ C2 ☐ C6 ☐ C7		
10					☐ C1 ☐ C2 ☐ C6 ☐ C7		
11					☐ C1 ☐ C2 ☐ C6 ☐ C7		
12					☐ C1 ☐ C2 ☐ C6 ☐ C7		
13					☐ C1 ☐ C2 ☐ C6 ☐ C7		
14					☐ C1 ☐ C2 ☐ C6 ☐ C7		
15					☐ C1 ☐ C2 ☐ C6 ☐ C7		
16					☐ C1 ☐ C2 ☐ C6 ☐ C7		
17					☐ C1 ☐ C2 ☐ C6 ☐ C7		
18					☐ C1 ☐ C2 ☐ C6 ☐ C7		
19					☐ C1 ☐ C2 ☐ C6 ☐ C7		

SET-UP



Null leads first · same reference riser for the area · compare risers (IEEE 81-2012 Cl. 10)

HOW TO RECORD

- Null the leads first; write lead R in the header (IEEE 81-2012 Cl. 10.2).
- Reading ☐ from the measurement point to the equipment body; reading ☐ from the same point to the grid / reference riser.
- Use one reference riser for the whole area so risers can be compared with each other.

Condition codes: C1 Corroded · C2 Strip / test link disconnected · C3 Single earth connection (optional) · C4 Buried / chamber not accessible or damaged · C5 Electrode damaged C6 Loose connection · C7 Joint not cleaned / tinned / protected · C8 No tag / label · OTH Other – write remark

ETPL engineer – name & signature	Client representative – name & signature	Time start / finish

MS-07 Ground (Grid) Resistance – Measurement Sheet

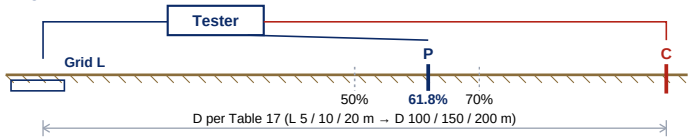
61.8% / slope / clamp · IS 3043:2018 Cl. 41, Table 17 · IEEE 81-2012 Cl. 8

Client Gujarat Fluorochemicals Ltd	**Project No.** 024-D03_TS-26-GFL	**Work Order No.** 1002-1261201769
Site / Unit Dahej – FI Plant	**Work date** ____ / ____ / 20____	**Sheet** ____ of ____
ETPL engineer(s)	**Client representative**	**Ground condition** ☐ Dry ☐ Damp ☐ Wet
Max. grid dimension L (m)	**Table 17 distance C used (m)**	**90°/180° check** ☐ Yes ☐ No

Instrument used: ☐ Metrel MI 3290 GX Earth Analyser ☐ Kyoritsu 4105A Digital Earth Tester Sr. No. _____ Cal. due ___/___/____

Sr.	Point / pit TAG	Location of measured ground	Method	Current probe distance C (m)	R at 50% (Ω)	R at 61.8% (Ω)	R at 70% (Ω)	Condition codes	Remark
1			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	
2			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	
3			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	
4			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	
5			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	
6			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	
7			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	
8			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	
9			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	
10			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	
11			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	
12			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	
13			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	
14			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	
15			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	
16			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	
17			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	
18			☐ 61.8% ☐ Slope ☐ Clamp					☐ C1 ☐ C2 ☐ C6 ☐ C7	

SET-UP



Read at 61.8%, confirm at 50% and 70% – values must agree (IS 3043:2018 Cl. 41.1.6)

HOW TO RECORD

- Grid stays connected (in-service value). Choose C from Table 17 by the grid's maximum dimension L.
- The 61.8% value is valid when the 50% and 70% readings agree; if not, move C further away or use the slope method (Cl. 41.1.8).
- Clamp method: no auxiliary probes – write '—' in the distance and 50/70% columns.

Condition codes: C1 Corroded · C2 Strip / test link disconnected · C3 Single earth connection (optional) · C4 Buried / chamber not accessible or damaged · C5 Electrode damaged · C6 Loose connection · C7 Joint not cleaned / tinned / protected · C8 No tag / label · OTH Other – write remark

ETPL engineer – name & signature	Client representative – name & signature	Time start / finish

MS-08 Step & Touch Voltage – Measurement Sheet

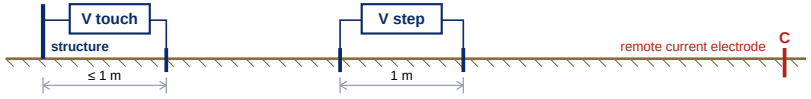
Current injection · IEEE 81-2012 Cl. 9 · IS 3043:2018 Cl. 10.3 · IEEE 80-2013

Client Gujarat Fluorochemicals Ltd	**Project No.** 024-D03_TS-26-GFL	**Work Order No.** 1002-1261201769
Site / Unit Dahej – FI Plant	**Work date** ____ / ____ / 20____	**Sheet** ____ of ____
ETPL engineer(s)	**Client representative**	**Ground condition** ☐ Dry ☐ Damp ☐ Wet
Earth fault current If (kA) & source	**Fault clearing time ts (s) & source**	**Remote current electrode distance (m)**
Surface layer ρs (Ω·m) & thickness hs (m)	**Body weight basis** ☐ 50 kg ☐ 70 kg ☐ Both	**Injection point**

Instrument used: ☐ Metrel MI 3295 Step & Contact Voltage Meter* ☐ Metrel MI 3290 GX Earth Analyser Sr. No. _____ Cal. due ___/___/____

Sr.	Location / structure / measurement point	Measurement type	Surface	Distance from structure (m)	Test current It (A)	V1 (V)	V2 (V)	V3 (V)	Vmax (V)	Photo ref.	Remark
1		☐ Step ☐ Touch-structure ☐ Touch-mesh ☐ Touch-fence ☐ Touch-gate	☐ Soil ☐ Gravel ☐ Concrete								
2		☐ Step ☐ Touch-structure ☐ Touch-mesh ☐ Touch-fence ☐ Touch-gate	☐ Soil ☐ Gravel ☐ Concrete								
3		☐ Step ☐ Touch-structure ☐ Touch-mesh ☐ Touch-fence ☐ Touch-gate	☐ Soil ☐ Gravel ☐ Concrete								
4		☐ Step ☐ Touch-structure ☐ Touch-mesh ☐ Touch-fence ☐ Touch-gate	☐ Soil ☐ Gravel ☐ Concrete								
5		☐ Step ☐ Touch-structure ☐ Touch-mesh ☐ Touch-fence ☐ Touch-gate	☐ Soil ☐ Gravel ☐ Concrete								
6		☐ Step ☐ Touch-structure ☐ Touch-mesh ☐ Touch-fence ☐ Touch-gate	☐ Soil ☐ Gravel ☐ Concrete								
7		☐ Step ☐ Touch-structure ☐ Touch-mesh ☐ Touch-fence ☐ Touch-gate	☐ Soil ☐ Gravel ☐ Concrete								
8		☐ Step ☐ Touch-structure ☐ Touch-mesh ☐ Touch-fence ☐ Touch-gate	☐ Soil ☐ Gravel ☐ Concrete								
9		☐ Step ☐ Touch-structure ☐ Touch-mesh ☐ Touch-fence ☐ Touch-gate	☐ Soil ☐ Gravel ☐ Concrete								
10		☐ Step ☐ Touch-structure ☐ Touch-mesh ☐ Touch-fence ☐ Touch-gate	☐ Soil ☐ Gravel ☐ Concrete								
11		☐ Step ☐ Touch-structure ☐ Touch-mesh ☐ Touch-fence ☐ Touch-gate	☐ Soil ☐ Gravel ☐ Concrete								
12		☐ Step ☐ Touch-structure ☐ Touch-mesh ☐ Touch-fence ☐ Touch-gate	☐ Soil ☐ Gravel ☐ Concrete								

SET-UP



Record It per reading · V actual = V measured × If / It → compare with tolerable limit (IEEE 81-2012 Cl. 9.3)

HOW TO RECORD

- Step: two probes 1 m apart. Touch: probe within 1 m of the structure (arm reach). Probes ~150 mm into moist soil (IEEE 81-2012 Cl. 9.2–9.3).
- Write the test current It for every row – the app scales each voltage by If / It before comparing with the tolerable limit.
- *MI 3295 is not on the current instrument list – confirm availability.

ETPL engineer – name & signature	Client representative – name & signature	Time start / finish

MS-09 Tower Footing Impedance – Measurement Sheet

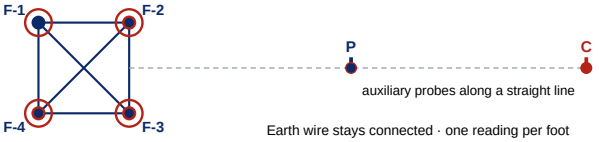
Clamp method · IS 3043:2018 Cl. 41 · IEEE 81-2012 Cl. 8

Client Gujarat Fluorochemicals Ltd	**Project No.** 024-D03_TS-26-GFL	**Work Order No.** 1002-1261201769
Site / Unit Dahej – FI Plant	**Work date** ____ / ____ / 20____	**Sheet** ____ of ____
ETPL engineer(s)	**Client representative**	**Ground condition** ☐ Dry ☐ Damp ☐ Wet
Line / circuit name	**Line voltage (kV) / tower type**	**Zsat (Ω) from acceptance list**

Instrument used: ☐ Metrel MI 3290 GX Earth Analyser Sr. No. _____ Cal. due ___/___/____

Sr.	Tower no. / location	Foot	Foot-to-earth connection	Measured current I (mA)	Measured impedance Z (Ω)	Condition codes	Photo ref.	Remark
1		F-1	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
		F-2	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
		F-3	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
		F-4	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
2		F-1	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
		F-2	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
		F-3	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
		F-4	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
3		F-1	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
		F-2	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
		F-3	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
		F-4	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
4		F-1	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
		F-2	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
		F-3	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
		F-4	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
5		F-1	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
		F-2	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
		F-3	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		
		F-4	☐ Given ☐ Not given			☐ C1 ☐ C2 ☐ C5 ☐ C6		

SET-UP



plan view – clamp (red) on each foot earth connection

Earth wire stays connected · one reading per foot

Zt = Σ 4 feet vs Zsat (acceptance list)

HOW TO RECORD

- Four feet per tower are one unit. Clamp each foot's earth connection in turn; the overhead earth wire stays connected.
- Mark 'Not given' where a foot has no earth connection – that foot is reported as a defect.
- The app calculates Zt from the four feet and compares it with Zsat from the project's acceptance profile.

Condition codes: C1 Corroded · C2 Strip / test link disconnected · C3 Single earth connection (optional) · C4 Buried / chamber not accessible or damaged · C5 Electrode damaged · C6 Loose connection · C7 Joint not cleaned / tinned / protected · C8 No tag / label · OTH Other – write remark

ETPL engineer – name & signature	Client representative – name & signature	Time start / finish